Program Evaluation with Observational Data

Chapter 3 - The LATE Model

Advanced Empirical Research Method

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Quote of the Day

“Theories are testable where they are least needed, and are not testable where they are most needed.” -Charles Manski

Instrumental Variables with Homogeneous Effect

Running example:
Is there an effect of military service on subsequent labour market outcomes?

$$\text{Wage} = \alpha + \beta \text{Service} + \gamma \textit{Ability} + \varepsilon$$

Instrumental Variable: Requires relevance, Exogeneity/relevance.

- **relevance**: the instrument actually changes the treatment;
- **Exogeneity**: the instrument does not change the outcome

Instrumental Variables with Homogeneous Effect

More Formally:

$$\beta_{OLS} = \frac{\text{Cov}(D, Y)}{\text{Var}(D)} = \frac{\beta \text{Cov}(D, D) + \text{Cov}(D, U)}{\text{Var}(D)} = \beta + \frac{\text{Cov}(D, U)}{\text{Var}(D)}$$

which does not equal β unless $\text{Cov}(D, U) = 0$.

$$\beta_{IV} = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)} = \frac{\beta \text{Cov}(Z, D) + \text{Cov}(Z, U)}{\text{Cov}(Z, D)} = \beta + \frac{\text{Cov}(Z, U)}{\text{Cov}(Z, D)} = \beta.$$

β_{IV} is unbiased if $\text{Cov}(Z, U) = 0$ (Exogeneity) and $\text{Cov}(Z, D) \neq 0$ (Relevance).

IV: Wald Estimand

When D and Z are both binary, the IV estimand can be written:

$$\beta_{IV} = \frac{p(1-p)\left[\mathbb{E}(Y|Z=1) - \mathbb{E}(Y|Z=0)\right]}{p(1-p)\left[\mathbb{E}(D|Z=1) - \mathbb{E}(D|Z=0)\right]} = \frac{\mathbb{E}(Y|Z=1) - \mathbb{E}(Y|Z=0)}{\mathbb{E}(D|Z=1) - \mathbb{E}(D|Z=0)} \tag{1}$$

where $p \equiv \mathbb{P}(Z=1)$.

Critical:
This is a homogeneous treatment effects model: treatment effect is the same for everyone.

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- Assumptions
- Identification result

How useful is the LATE?

LATE: Generalisation and Estimation
- Multiple instruments
- Covariates

Extension: Testing LATE asumptions

Running Example

Angrist (1990):

- Causal relationship of interest: The effects of the Vietnam-era military service on the earnings of the veterans.
- Selection problem: Veterans are not a random sample of the population.
- Identification: Draft-eligibility lottery as instrument.
  "In the 1960s and 1970s, young men were at risk of being drafter for military service. Concerns about the fairness of US conscription policy led to the institution of a draft lottery [over birthdays] that was used to determine priority for conscription."

Running Example

Angrist (1990):

- Random sequence numbers were randomly assigned to each birth date.
- Ceiling below which men were eligible for the draft.
- Men with large numbers could not be drafted.
- Many draft-eligible were still exempted from service for health and other reasons.
- Many men who were draft-exempt volunteered.

Concern: decision to comply to draft-eligibility is correlated with the treatment effect intensity.

Non-Compliance

Example: Effect of veteran status on income

"Educational draft deferments would have led men with low lottery numbers to stay in college longer than they would have otherwise desired. If so, draft lottery numbers are correlated with earnings ***for at least two reasons****: an increased likelihood of military service and an increased likelihood of college attendance."*

"The fact that the lottery number is randomly assigned (and therefore satisfies the independence assumption) ***does not*** *make this possibility less likely."*

Non-Compliance

" In the example of Vietnam veterans, the instrument (the draft lottery number) fails to be exogenous because the error term in the earnings equation ***depends on each individual's rate of return*** *to schooling, and* ***whether or not potential draftee accepted*** *their assignment [...] depends on the rate of return."* Deaton, 2010.

Lesson:
The effect identified by IV depends on *compliance* with draft eligibility.

Heterogeneity in Treatment Effect

Let Y be the earnings of i, D: veteran status, and Z: the randomized draft-eligibility.

$$\begin{aligned} Y &= Y_0 + (Y_1 - Y_0)D \\ &= \alpha + \beta D + u. \end{aligned}$$

where

$$\begin{aligned} \alpha &= E(Y_0), \\ \beta &= \mathbb{E}(Y_1 - Y_0 | D = 1), \text{ and} \\ u &= Y_0 - \mathbb{E}(Y_0) + [(Y_1 - Y_0) - \mathbb{E}(Y_1 - Y_0 | D = 1)]\, D \\ &:= v + (\tau - \bar{\tau})D \end{aligned}$$

Heterogeneity in Treatment Effect

Condition for consistency: $\mathbb{E}(Zu) = 0$

$$\begin{aligned}
\mathbb{E}(Zu) &= \mathbb{E}(Z(v + (\tau - \bar{\tau})D)) \\
&= \mathbb{E}(Z(\tau - \bar{\tau})D), \text{ by random assignment} \\
&= \mathbb{E}(\tau - \bar{\tau} | Z = 1, D = 1) \times P(Z = 1, D = 1)
\end{aligned}$$

The exogeneity condition holds if:

1. The average effect of military service among the veterans induced into military service by the draft-eligibility is the same as the average effect among those who would have participated anyway $\mathbb{E}(Y_1 - Y_0 | Z = 1, D = 1) - \mathbb{E}(Y_1 - Y_0 | D = 1)$, (a stronger restriction is: $\tau = \bar{\tau}$).
2. or no one who is drafted enrolls $P(Z = 1, D = 1) = 0$.

IV Estimand in Random Coefficient Model

How can we interpret the IV estimand?

Consider the random coefficients linear IV model:

$$\begin{aligned} Y &= \alpha + BD + U \\ D &= \pi + CZ + V \end{aligned}$$

with α, π are constant, B, C, U, V are unobservable random variables.

More specifically: $U = Y_0 - \mathbb{E}(Y_0)$ and $B = Y_1 - Y_0$ (the individual-specific TE).

Assume $Z \perp\!\!\!\perp (U, V, B, C)$ (exogeneity) and $\text{Cov}(D, Z) \neq 0$ (Relevance). Then the IV estimand is given by:

$$\beta_{IV} = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)}$$

IV Estimand in Random Coefficient Model

One can show that:

$$\beta_{IV} = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)} = \mathbb{E}\left[\frac{C}{\mathbb{E}(C)} B\right] = \mathbb{E}[\omega B]$$

1. β_{IV} is **weighted average** of the causal effect of D on Y.
2. When B is constant, $\beta_{IV} = B$.
3. Agents more strongly impacted by Z have a larger $|C|$, and receive a higher weight.

IV Estimand in Random Coefficient Model

One can show that:

$$\beta_{IV} = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)} = \mathbb{E}\left[\frac{C}{\mathbb{E}(C)} B\right] = \mathbb{E}[\omega B]$$

β_{IV} is difficult to interpret - unlikely to be representative of the overall population:

1. Overweights TE for agents more affected by the instruments ;
2. β_{IV} is different for different instruments;
3. Weights can be negative for those with $C \times \mathbb{E}(C) < 0$, so that $\beta_{IV} \approx 0$ although $B > 0$.

LATE Model

Imbens and Angrist (1994):

- The paper provides additional conditions under which β_{IV} is easier to interpret;
- Assumptions:
  1. Exclusion + Exogeneity - in two parts, stronger than the usual linear IV ;
  2. Relevance - as usual;
  3. Monotonicity - A new condition that ensures that we don't face 'negative weights'.
- Under these conditions: β_{IV} is the TE for the so-called **compliers**.

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Extension: Testing LATE asumptions

Compliance Types

Mathematically, we can list every function that maps Z to D.

We give each a name corresponding to the "type" (T) of draftee:

$$\begin{aligned}
\text{Never-taker:} \quad & T = n & \iff \quad & D(Z) = 0 \\
\text{Always-taker:} \quad & T = a & \iff \quad & D(Z) = 1 \\
\text{Complier:} \quad & T = c & \iff \quad & D(Z) = Z \\
\text{Defier:} \quad & T = d & \iff \quad & D(Z) = (1 - Z).
\end{aligned}$$

Compliance Types

Mathematically, we can list every function that maps Z to D.

We give each a name corresponding to the "type" (T) of draftee:

$$\begin{aligned}
\text{Never-taker:} \quad & T = n & \iff \quad & D(1) = D(0) = 0 \\
\text{Always-taker:} \quad & T = a & \iff \quad & D(1) = D(0) = 1 \\
\text{Complier:} \quad & T = c & \iff \quad & D(1) > D(0) \\
\text{Defier:} \quad & T = d & \iff \quad & D(1) < D(0).
\end{aligned}$$

LATE Model Assumptions: Unconfounded Types

Assumption (Unconfounded Type)

For all compliance types $t \in \{a, c, n, d\}$

$$\mathbb{P}(T = t) = \mathbb{P}(T = t | Z = 0) = \mathbb{P}(T = t | Z = 1).$$

Interpretation:
Knowing the value of a person's instrumental variable tells us nothing about her type. It holds automatically if Z is randomly assigned.

LATE Model Assumptions: Monotonicity

Assumption (No Defiers / Monotonicity)

$\mathbb{P}(T = d) = 0.$

Interpretation:
The instrument shifts everyone in the *same direction* or not at all.

Question

Do you think it is the case with the drafter?

LATE Model Assumptions

Lemma

Under Unconfounded Types and Monotonicity:

$$\mathbb{P}(D=1|Z=1)=\mathbb{P}(T=a)+\mathbb{P}(T=c)$$
$$\mathbb{P}(D=0|Z=1)=\mathbb{P}(T=n)$$
$$\mathbb{P}(D=1|Z=0)=\mathbb{P}(T=a)$$
$$\mathbb{P}(D=0|Z=0)=\mathbb{P}(T=c)+\mathbb{P}(T=n)$$

For proof: See lecture Notes (Core ERM).

LATE Model Assumptions

Lemma

Under Unconfounded Type and Monotonicity:

$$\mathbb{P}(D=1|Z=1)=\mathbb{P}(T=a)+\mathbb{P}(T=c)$$
$$\mathbb{P}(D=0|Z=1)=\mathbb{P}(T=n)$$
$$\mathbb{P}(D=1|Z=0)=\mathbb{P}(T=a)$$
$$\mathbb{P}(D=0|Z=0)=\mathbb{P}(T=c)+\mathbb{P}(T=n)$$

Implication:

To obtain the share of compliers, we take differences:

$$\mathbb{P}(T=c)=\mathbb{P}(D=1|Z=1)-\mathbb{P}(D=1|Z=0).$$

LATE Model Assumptions: Exclusion

Weaker form:

Assumption (Mean Exclusion)

$\mathbb{E}[Y_d|Z = z, T = t] = \mathbb{E}[Y_d|T = t]$ for all (d, z, t).

Stronger form (distribution):

Assumption (Exclusion)

$F_{Y_d}[y|Z = z, T = t] = F_{Y_d}[y|T = t]$ for all (d, z, t).

LATE Model Assumptions: Mean Exclusion

Lemma

Under Assumptions unconfounded types, no-defiers, and mean exclusion:

$$\mathbb{E}[Y|D=1,Z=1] = \frac{\mathbb{P}(T=a)\mathbb{E}[Y_1|T=a]+\mathbb{P}(T=c)\mathbb{E}[Y_1|T=c]}{\mathbb{P}(T=a)+\mathbb{P}(T=c)}$$

$$\mathbb{E}[Y|D=0,Z=1] = \mathbb{E}[Y_0|T=n]$$

$$\mathbb{E}[Y|D=1,Z=0] = \mathbb{E}[Y_1|T=a]$$

$$\mathbb{E}[Y|D=0,Z=0] = \frac{\mathbb{P}(T=n)\mathbb{E}[Y_0|T=n]+\mathbb{P}(T=c)\mathbb{E}[Y_0|T=c]}{\mathbb{P}(T=n)+\mathbb{P}(T=c)}$$

Question

Proof: See the lectures Notes Core ERM.

LATE Model Assumptions: First-stage

Assumption (Existence of Compliers)

$\mathbb{P}(T = c) > 0$

Interpretation

The instrument shifts the compliance decision of at least part of the population.

$$\mathbb{E}(D|Z = 1) > \mathbb{E}(D|Z = 0)$$

LATE Theorem

Theorem

Under Assumptions unconfounded types, no-defiers, mean exclusion, and existence of compliers,

$$\frac{\mathbb{E}(Y|Z=1)-\mathbb{E}(Y|Z=0)}{\mathbb{E}(D|Z=1)-\mathbb{E}(D|Z=0)}=\mathbb{E}\left[Y_1-Y_0|\,T=c\right]$$

Interpretation

IV does not in general identify the ATE in a heterogenous treatment effects setting. The Wald estimand identifies the Average Treatment Effect for the Compliers.

Proof: See the lectures Notes Core ERM.

LATE Theorem

Interpretation:

"...an instrument which is as good as randomly assigned, affects the outcome through a single known channel, has a first-stage, and affects the causal channel of interest only in one direction, can be used to estimate the average causal effect on the ***affected group****."*

"[IV] estimate[s] the effect of military service on men who served ***because they were draft-eligible****, but would not otherwise have served."*

IV and Causality: Wald Estimator

Angrist (1990):

Earnings year	Earnings: Mean (1)	Earnings: Eligibility Effect (2)	Veteran Status: Mean (3)	Veteran Status: Eligibility Effect (4)	Wald Estimate of Veteran Effect (5)
1981	16,461	-435.8	0.267	0.159	-2,741
		(210.5)		(0.040)	(1,324)
1971	3,338	-325.9			-2050
		(46.6)			(293)
1969	2,299	-2.0			
		(34.5)			

Why Monotonicity?

Note that

$$\mathbb{E}(Y|Z=1) = \mathbb{E}(Y_0 + (Y_1 - Y_0)D_1|Z=1)$$
$$\mathbb{E}(Y|Z=0) = \mathbb{E}(Y_0 + (Y_1 - Y_0)D_0|Z=0)$$

Take the difference:

$$\begin{aligned}
\mathbb{E}(Y|Z=1) - \mathbb{E}(Y|Z=0) &= \mathbb{E}((Y_1 - Y_0)(D(1) - D(0))) \\
&= \mathbb{E}(Y_1 - Y_0|D(1) > D(0))\,P(D(1) > D(0)) \\
&\quad -\mathbb{E}(Y_1 - Y_0|D(1) < D(0))\,P(D(1) < D(0)) \\
&= \mathbb{E}(Y_1 - Y_0|D(1) > D(0))\,P(D(1) > D(0)) \\
&\quad -\mathbb{E}(Y_1 - Y_0|D(1) < D(0))\,P(D(1) < D(0))
\end{aligned}$$

Heterogeneity in Treatment Effect

When the treatment effect is not constant:

$$\begin{aligned} \mathbb{E}\left(Y_1 - Y_0 | D(1) > D(0)\right) &> 0 \\ \mathbb{E}\left(Y_1 - Y_0 | D(1) < D(0)\right) &> 0 \end{aligned}$$

does not guarantee that the Wald estimator is positive.

Treatment effect can be positive for everyone, yet, the reduced form can be zero or even negative. The negative weights in the expression β_{IV} are the result of this.

To solve the problem, LATE assumes either $P\left(D(1) < D(0)\right) = 0$ or $P\left(D(1) > D(0)\right) = 0$

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How useful is the LATE?

1. Compliers are specific to the instrument: this raises the question of how generalisable the insight from the LATE are.
2. Special cases where $LATE = ATT$ or $LATE = ATNT$: In these special cases, the LATE is a causal effect of interest.
3. Counting and characterizing the "compliers": if one can show who the compliers are, that they form a 'representative' part of the population, then there is some room for extrapolation.

LATE: effect identified

LATE is not informative about effects on never-takers and always-takers because, by definition, treatment status for these two groups is unchanged by the instrument.

In general: $ATE \neq LATE$, $ATT \neq LATE$, $ATNT \neq LATE$:

- ATE: whole population;
- ATT: compliers (with instrument switched on) and always-takers;
- $ATNT$: compliers (with instrument switched off) and never-takers;

LATE: effect identified

ATT:

$$
\begin{aligned}
\mathbb{E}(Y_1 - Y_0 | D = 1) &= \mathbb{E}(Y_1 - Y_0 | D(0) = 1, D = 1)\, P(D(0) = 1 | D = 1) \\
&\quad + \mathbb{E}(Y_1 - Y_0 | D(1) > D(0), Z = 1)\, P(D(1) > D(0), Z = 1 | D = 1) \\
&= \underbrace{\mathbb{E}(Y_1 - Y_0 | D(0) = 1)}_{\text{effect on always-takers}} P(D(0) = 1 | D = 1) \\
&\quad + \underbrace{\mathbb{E}(Y_1 - Y_0 | D(1) > D(0))}_{\text{effect on compliers}} P(D(1) > D(0), Z = 1 | D = 1)
\end{aligned}
$$

LATE: effect identified

ATNT:

$$\mathbb{E}(Y_1 - Y_0 | D = 0) = \underbrace{\mathbb{E}(Y_1 - Y_0 | D(1) = 0)}_{\text{effect on never-takers}} P(D(0) = 0 | D = 0) + \underbrace{\mathbb{E}(Y_1 - Y_0 | D(1) > D(0))}_{\text{effect on compliers}} P(D(1) > D(0), Z = 0 | D = 0)$$

ATE:

$$\mathbb{E}(Y_1 - Y_0) = \mathbb{E}(Y_1 - Y_0 | D = 1) P(D = 1) + \mathbb{E}(Y_1 - Y_0 | D = 0) P(D = 0).$$

LATE: Special cases

Twin instrument (Angrist and Evans, 1998):

Causal relationship of interest: Effect of fertility on labor force participation.

Treatment D : Third birth.
Instrument Z: Multiple second birth.

Assumptions:

- Z is randomly assigned,
- Multiple births affect outcomes only by increasing fertility,
- No one has a lower fertility because of a multiple birth.

LATE: Special cases

Twin instrument (Angrist and Evans, 1998):

$$LATE = ATNT = \mathbb{E}(Y_1 - Y_0 | D = 0)$$

"all women who have a multiple second birth end up with three children, i.e., there are ***no never-takers*** *in response to the twins instrument."*

LATE: Special cases

Angrist and Evans (1998):

Table 4.1.4: Wald estimates of labor supply effects

			IV Estimates using:			
			Twins		Sex-composition	
Dependent variable	Mean (1)	OLS (2)	First stage (3)	Wald estimates (4)	First stage (5)	Wald estimates (6)
Employment	0.528	-0.167	0.625	-0.083	0.067	-0.135
		(0.002)	(0.011)	(0.017)	(0.002)	(0.029)
Weeks worked	19.0	-8.05	"	-3.83	"	-6.23
		(0.09)	"	(0.758)	"	(1.29)
Hours/week	16.7	-6.02	"	-3.39	"	-5.54
		(0.08)	"	(0.637)	"	(1.08)

LATE: effect identified

Chattopadhyay and Duflo (2004):

Causal relationship of interest: Effect of women as local ruler on development outcome.

Treatment D : women as a local chief.
Instrument Z: reservation seat. Assumptions:

- Z is randomly assigned,
- Reservation outcomes only through the woman position,
- **No village would have man local chief** because the seat was reserved to a women.

LATE: Special Cases

Chattopadhyay and Duflo (2004):

TABLE I

FRACTION OF WOMEN AMONG PRADHANS IN RESERVED AND UNRESERVED GP

	Reserved GP (1)	Unreserved GP (2)
West Bengal		
Total Number	54	107
Proportion of Female Pradhans	100%	6.5%
Rajasthan		
Total Number	40	60
Proportion of Female Pradhans	100%	1.7%

LATE: Special Cases

Randomized trials with one sided non-compliance.

Causal relationship of interest: Effect of treatment on outcome.

Treatment D : treatment.
Instrument Z: randomized treatment assignment.

One sided non-compliance:

- Randomized trial.
- Voluntary participation induces imperfect compliance.
- **No one in the control group** has access to the treatment, $\mathbb{E}(D|Z=0)=0$.

LATE: Special Cases

Randomized trials with one sided non-compliance.

$$
\begin{aligned}
LATE = ATT &= \frac{ITT}{\text{Compliance rate}} \\
&= \frac{\mathbb{E}(Y|Z=1) - \mathbb{E}(Y|Z=0)}{\mathbb{E}(D|Z=1)}
\end{aligned}
$$

LATE: Counting and Characterizing the Compliers

How representative are compliers of the whole population?

Proportion of compliers:

$$P(D(1) > D(0)) = \mathbb{E}(D|Z = 1) - \mathbb{E}(D|Z = 0)$$

Proportion of compliers among treated:

$$\begin{aligned} P(D(1) > D(0)|D = 1) &= \frac{P(D = 1|D(1) > D(0))P(D(1) > D(0))}{P(D = 1)} \\ &= \frac{P(Z = 1)(\mathbb{E}(D|Z = 1) - \mathbb{E}(D|Z = 0))}{P(D = 1)} \end{aligned}$$

LATE: Counting and Characterizing the Compliers

Endogenous Variable (D) (2)	Instrument (Z) (3)	Sample (4)	$P[D=1]$ (5)	1st Stage, $P[D_1 > D_0]$ (6)	$P[Z=1]$ (7)	$P[D_1 > D_0 \mid D=1]$ (8)	$P[D_1 > D_0 \mid D=0]$ (9)
Veteran Status	Draft eligibility	White men born in 1950	0.267	0.159	0.534	0.318	0.101
		Non-white men born in 1950	0.163	0.060	0.534	0.197	0.033
More than 2 children	Twins at second birth	Married women aged 21-35 with two or more children in 1980	0.381	0.603	0.008	0.013	0.966
	First two children are of the same sex	Married women aged 21-35 with two or more children in 1980	0.381	0.060	0.506	0.080	0.048
High school graduate	Third or fourth quarter birth	Men born between 1930 and 1939	0.770	0.016	0.509	0.011	0.034

LATE: Counting and Characterizing the Compliers

How representative are compliers of the whole population?

Proportion of individuals with characteristics X among compliers compared to the proportion in the population:

$$\begin{aligned}\frac{P(X=1|D(1)>D(0))}{P(X=1)} &= \frac{P(D(1)>D(0)|X=1)}{P(D(1)>D(0))} \\ &= \frac{(\mathbb{E}(D|Z=1,X=1)-\mathbb{E}(D|Z=0,X=1))}{(\mathbb{E}(D|Z=1)-\mathbb{E}(D|Z=0))}\end{aligned}$$

LATE: Counting and Characterizing the Compliers

Variable	$E[x]$ (1)	Twins at second birth: $E[x \mid D_1 > D_0]$ (2)	Twins at second birth: $P[x \mid D_1 > D_0] / P[X]$ (3)
Age 30 or older at first birth	0.00291	0.00404	1.39 (0.0201)
Black or hispanic	0.125	0.103	0.822 (0.00421)
High school graduate	0.822	0.861	1.048 (0.000772)
College graduate	0.132	0.151	1.14 (0.00376)

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LATE: Multiple instruments

Suppose $Z \in \{0, 1, \ldots, L\}$ with $L \geq 2$. The LATE assumption can be adjusted as follows:

1. Exogeneity: $(Y_0, Y_1, D(0), D(1), \ldots, D(L)) \perp\!\!\!\perp Z | X$;
2. Relevance: $P(D = 1 | Z = z) \neq P(D = 1 | Z = z')$ for at least some $z' \geq z$;
3. Monotonicity: $D(z') \geq D(z)$ (or conversely) for all $z' \geq z$.

Furthermore, one can order the instrument. Note that by exogeneity: $p(z) = P(D_z = 1)$. Under monotonicity, $D(z') \geq D(z)$ if and only if $p(z') \geq p(z)$. So we can relabel Z such that:

$$p(z_0) \leq p(z_1) \leq \ldots \leq p(z_L)$$

LATE: Multiple instruments

Consider the TSLS estimator using $\{I\{Z=z\} : z=0,1,\ldots,L\}$ as instruments. The slope of coefficient estimand on D is given by:

$$\beta_{TSLS} = \frac{\mathrm{Cov}\left(Y, p(Z)\right)}{\mathrm{Cov}\left(D, p(Z)\right)}$$

It is possible to show that:

$$\beta_{TSLS} = \sum_{m=1} \lambda_m LATE^m_{m-1}, \text{ where } LATE^m_{m-1} := E(Y_1 - Y_0 | D_m = 1, D_{m-1} = 0)$$

$$\text{and } \lambda_m := \frac{\left(p(z_m) - p(z_{m-1})\right)\left(\sum_{l=m}^{L}\left[p(z_l) - E\left(p(Z)\right)\right]P(Z=z_l)\right)}{\sum_{n=1}^{L}\left(p(z_n) - p(z_{n-1})\right)\left(\sum_{l=n}^{L}\left[p(z_l) - E\left(p(Z)\right)\right]P(Z=z_l)\right)} > 0$$

LATE: Multiple instruments

The TSLS is a **weighted average** of **pairwise LATEs**:

$$\beta_{TSLS} = \sum_{m=1} \lambda_m LATE^m_{m-1},$$

The LATEs in the average are for compliers at $Z = m-1$ to $Z = m$: individuals who switch treatment status because the instruments has moved from $m-1$ to m.

Note that the weights are **all positive** and **sum to 1**.

Unfortunately, these nice properties **do not** carry on when one introduces covariates (Blandhol et al., 2022).

LATE: Covariates

The introduction of covariates serves three purposes:

1. Justify exogeneity of Z;
2. Reduce residual noise in Y;
3. Detect observed heterogeneity in treatment effects.

Then, one can adjust the LATE assumptions as follows:

1. Exogeneity: $(Y_0, Y_1, D(0), D(1)) \perp\!\!\!\perp Z|X$;
2. Relevance: $P(D=1|X, Z=1) \neq P(D=1|X, Z=0)$ a.s. ;
3. Monotonicity: $P(D(0) \leq D(1)|X) = 1$ a.s. ;
4. Overlap: $P(Z=1|X) \in (0,1)$ a.s. ;

LATE: Estimation

With the same argument as before, one can show that:

$$E(Y_1 - Y_0 | T = c, X = x) = LATE(x)$$

And one can aggregate the effects to obtain:

$$E(Y_1 - Y_0 | T = c) = E\left[\frac{LATE(X)P(T = c|X)}{P(T = c)}\right]$$

The standard approach is to conduct a Two-Stage Least Square regression (TSLS). But the weights are not guaranteed to be positive. Difficult to interpret for counterfactual!

LATE: Estimation

Frölich (2007) suggests three estimators in the case with covariates, where D and Z are binary. Define:

$\widehat{m}_d(x) = \widehat{E}(Y|X = x, Z = d)$: a nonparametric estimator of the conditional moment,
$\widehat{\mu}_d(x) = \widehat{E}(D|X = x, Z = d)$: a nonparametric estimator of the propensity score.

The ATE of the population of compliers is nonparametrically estimated by

1. A ratio between two *matching estimators*:

$$\widehat{E}(Y_1 - Y_0|T = c) = \frac{\sum_{i:Z_i=1}\left(Y_i - \hat{m}_0(X_i)\right) - \sum_{i:Z_i=0}\left(Y_i - \widehat{m}_1(X_i)\right)}{\sum_{i:Z_i=1}\left(D_i - \widehat{\mu}_0(X_i)\right) - \sum_{i:Z_i=0}\left(D_i - \widehat{\mu}_1(X_i)\right)}$$

LATE: Estimation

Define:

$\widehat{\pi}(x) = \hat{E}(D|X = x)$: a nonparametric estimator of the propensity score.
$\widehat{m}_{\pi d}(\rho) = \hat{E}(D|\pi(X) = \rho, Z = d)$: an estimator of the conditional moment at a given propensity score.

2. An inverse propensity score weighting estimator:

$$\widehat{E}(Y_1 - Y_0|T = c) = \frac{\sum_i \left(Y_i Z_i/\widehat{\pi}(X_i) - Y_i(1 - Z_i)/(1 - \widehat{\pi}(X_i))\right)}{\sum_i \left(D_i Z_i/\widehat{\pi}(X_i) - D_i(1 - Z_i)/(1 - \widehat{\pi}(X_i))\right)}$$

3. A propensity score matching estimator (R package: causalweight):

$$\widehat{E}(Y_1 - Y_0|T = c) = \frac{\sum_i \left(\hat{m}_{\pi 1}(\hat{\pi}(X_i)) - \hat{m}_{\pi 0}(\hat{\pi}(X_i))\right)}{\sum_{i:Z_i=1} \left(\hat{\mu}_{\pi 1}(\hat{\pi}(X_i)) - \hat{\mu}_{\pi 0}(\hat{\pi}(X_i))\right)}$$

LATE: Estimation

Additional estimation alternatives:

- Fully parametric methods (Imbens and Rubin, 1997; Hirano et al., 2000; Yau and Little, 2001; Slocyzynski et al., 2022),
- Semiparametric methods (Abadie, 2003; Tan, 2006, 2010; Hong and Nekipelov, 2010),
- Nonparametric methods (Ogburn et al., 2015; Heiler, 2021; Sun and Tan, 2021), including machine learning based approaches (Chernozhukov et al., 2018; Athey et al., 2019; Singh and Sun, 2022).

See full references in Blandhol et al. (2022)
`https://a-torgovitsky.github.io/tsls-weights.pdf`.

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How useful is the LATE?

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- Covariates

Extension: Testing LATE asumptions

Testing LATE assumptions

They main result of Imbens and Angrist (1994) is that in presence of heterogenous treatment effect, we should be careful about the interpretation of the 2SLS estimand.

The 2SLS has a causal interpretation only under the LATE assumptions.

How realistic are those assumptions in empirical works? Are those assumptions testable?

There is a general belief in the applied-economics that the LATE assumptions are, in general, fundamentally non-testable.

Testing LATE assumptions

However, Balke and Pearl (1997, JASA) derive the first set of testable implications when Y, D, Z are all binary.

Heckman and Vytlacil (2005, ECA) generalized those testable implications to non-binary outcome and non-binary instrument.

Kitagawa (2015, ECA) highlighted the most informative restrictions that can screen out all the violations of the LATE assumptions when D and Z are binary and provides a statistical procedure implementable in R with the codes available under requests to the author.

Mourifié and Wan (2017, Restat) provides an alternative testing procedure using existing Stata packages.

LATE Testable implications: HV2005

The LATE Monotonicity assumption (LM) imposes: $D_0 \leq D_1$. Hence, for all $A \in \mathscr{B}_{\mathscr{Y}}$,

$$\begin{aligned}\mathbb{P}(Y \in A, D = 1 | Z = 0) &= \mathbb{P}(Y_1 \in A, D_0 = 1 | Z = 0) \\ &\overset{RA}{=} \mathbb{P}(Y_1 \in A, D_0 = 1) \overset{LM}{\leq} \mathbb{P}(Y_1 \in A, D_1 = 1) \\ &\overset{RA}{=} \mathbb{P}(Y_1 \in A, D_1 = 1 | Z = 1) \\ &= \mathbb{P}(Y \in A, D = 1 | Z = 1).\end{aligned}$$

In practice some applied papers look at the sign of the first stage to infirm or not if the LM holds which basically means $\mathbb{P}(D = 1 | Z = 0) \leq \mathbb{P}(D = 1 | Z = 1)$.

- They look at just the special case where $A = \mathscr{Y}$.
- This simple procedure has very low power in screening the violations of the LM.

Testable implication: HV2005

The Random Assignment assumption + the LATE Monotonicity assumption (RA + LM) together impose: $RA + LM \Rightarrow$ for all $A \in \mathscr{B}_{\mathscr{Y}}$

$$\begin{aligned} T_1(A) &\equiv \mathbb{P}(Y \in A, D = 1 | Z = 0) - \mathbb{P}(Y \in A, D = 1 | Z = 1) \leq 0, \text{ and} \\ T_0(A) &\equiv \mathbb{P}(Y \in A, D = 0 | Z = 1) - \mathbb{P}(Y \in A, D = 0 | Z = 0) \leq 0. \end{aligned}$$

Therefore for all $A \in \mathscr{B}_{\mathscr{Y}}$

$$T(A) = \max\{T_0(A), T_1(A)\} \leq 0. \text{ or, equivalently } \sup_{A \in \mathscr{B}_{\mathscr{Y}}} T(A) \leq 0.$$

Kitagawa (2015) testing procedure is directly based on this object.

Mourifié and Wan (2017) alternative representation.

Mourifié and Wan (2017, Restat, MW) provides an alternative testing procedure.

$$\text{Since} \quad \mathbb{P}(Y \in A, D = 1 | Z = 1) = \frac{\mathbb{P}(D = 1, Z = 1, Y \in A)}{\mathbb{P}(Z = 1)},$$

$$\mathbb{P}(Y \in A, D = 1 | Z = 0) = \frac{\mathbb{P}(D = 1, Z = 0, Y \in A)}{\mathbb{P}(Z = 0)},$$

$$\text{Then } T_1(A) \leq 0 \iff \frac{\mathbb{P}(D = 1, Z = 0, Y \in A)}{\mathbb{P}(Z = 0)} \leq \frac{\mathbb{P}(D = 1, Z = 1, Y \in A)}{\mathbb{P}(Z = 1)}$$

$$\iff \mathbb{E}\{\mathbf{1}_{Y \in A} \times [D(1 - Z)p_1 - DZp_0]\} \leq 0, \text{ where } p_j = \mathbb{P}(Z = j)$$

Mourifié and Wan (2017) alternative representation.

Therefore we have a.s.

$$\begin{cases} \theta(Y,1) \equiv \mathbb{E}[p_1 D(1-Z) - p_0 DZ | Y] \leq 0, \\ \theta(Y,0) \equiv \mathbb{E}[p_0 1(1-D)Z - p_1(1-D)(1-Z) | Y] \leq 0. \end{cases}$$

The outcome variable (Y) becomes a conditioning variable. This facilitates implementation: no need to visit all sets A. Furthermore, it is natural to incorporate X:

$$\begin{cases} \theta(Y,X,1) \equiv \mathbb{E}[p_1 D(1-Z) - p_0 DZ | Y,X] \leq 0, \\ \theta(Y,X,0) \equiv \mathbb{E}[p_0 1(1-D)Z - p_1(1-D)(1-Z) | Y,X] \leq 0. \end{cases}$$

Some empirical illustrations

- Mourifié and Wan kindly shared their empirical applications.
  - Angrist (1991), “Draft eligibility”
  - Card (1993), “college proximity”
  - Angrist and Evans (1998), “Same-sex”

“Draft eligibility”

- What is the effect of veteran status on civilian earnings?
- Endogeneity of the veteran status.
  - Certain types of men are more likely to serve in the armed forces than others
- Angrist (1991) proposes as instrument: A binary indicator of the draft eligibility.
  - Theoretically randomly assigned since based on ones birthdate through the draft lotteries.
  - Drafted are more likely to be Veteran?

"Draft eligibility": Data

Survey of Income and Program Participation (SIPP) 1984

Table: Summary Statistics of SIPP Data from Angrist (1991)

	Total	Draft Eligible	Veteran
Observations	3027	1379	994
Age	34.063 (2.804)	34.685 (2.607)	35.064 (2.494)
Veteran	0.328 (0.470)	0.403 (0.491)	1.000 (0.000)
Draft Eligible	0.456 (0.498)	1.000 (0.000)	0.560 (0.497)
Years of Schooling	13.522 (2.864)	13.578 (2.834)	13.443 (2.260)
Race(Non White)	0.118 (0.322)	0.116 (0.320)	0.080 (0.272)
log (Weekly Wage)	2.217 (0.532)	2.247 (0.534)	2.248 (0.498)

Average and standard deviation (in the parenthesis)

"Draft eligibility"

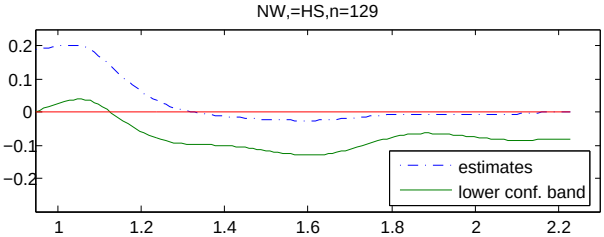
Table: Lottery local method (clrtest)

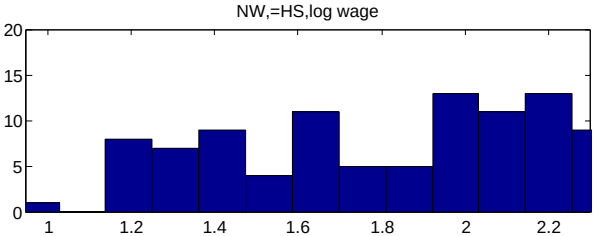
	W,<HS	W,=HS	W,>HS	NW,< 4S	NW,=HS	NW,>HS	All
Obs.	320	867	1482	56	129	173	3027
90%	NR	NR	NR	NR	R	NR	NR
95%	NR	NR	NR	NR	R	NR	NR
99%	NR	NR	NR	NR	NR	NR	NR

"R" stands for rejection and "NR" stands for no rejection.

- Default kernel: $K(u) = \frac{15}{16}(1-u^2)^2 \mathbf{1}\{|u| \leq 1\}$.
- Default bandwidth: $h_{ROT} \times \hat{s} \times n^{\frac{1}{5}} \times n^{-\frac{2}{7}}$, where h_{ROT} is given by Fan and Gijbels (1996).
- Grid: 2.5% quantile to 97.5% quantile of Y, step width 0.01.

“Draft eligibility”





“College proximity”

- What is the impact of the education on the mother labor income?
- Endogenous education level.
  - Correlated with unobserved ability
- Card (1993) proposes to use college proximity.
- A dummy for whether there is a college in the local labor market.
  - Birth place shall be independent with inherited ability
  - More job opportunities if there is college (e.g. someone who prefers to be a bartender?)

"College proximity"

Table: Summary Statistics of NLSYM Sample

	Total	$D = 1$	$Z = 1$
Observations	3005	2048	816
Lived in metro area in 1966	0.651 (0.476)	0.693 (0.461)	0.801(0.399)
Lived in southern states in 1966	0.414 (0.492)	0.313 (0.464)	0.329 (0.470)
Black	0.232 (0.422)	0.099 (0.299)	0.209 (0.407)
Years of Schooling in 1976	13.26 (2.675)	16.692 (0.849)	13.532 (2.577)
D (education $\geq$ 16)	0.271 (0.444)	1.000 (0.000)	0.293 (0.455)
Z (college proximity)	0.681 (0.465)	0.736 (0.015)	1.000 (0.000)
Y (log wage in 1976)	6.261 (0.444)	6.428 (0.433)	6.311 (0.440)

Average and standard deviation (in the parenthesis)

"College proximity"

Table: Subgroups of Card (1990)

	Non-Black (NB)	Black (B)
Non-Southern (NS) & Non-Metro (NM)	429	5
Non-Southern (NS) & Metro (M)	1191	138
Southern (S) & Non-Metro (NM)	307	314
Southern (S) & Metro (M)	380	246

Southern (south66): lived in southern states in 1966. Metro (smsa66r): lived in urban area in 1966.

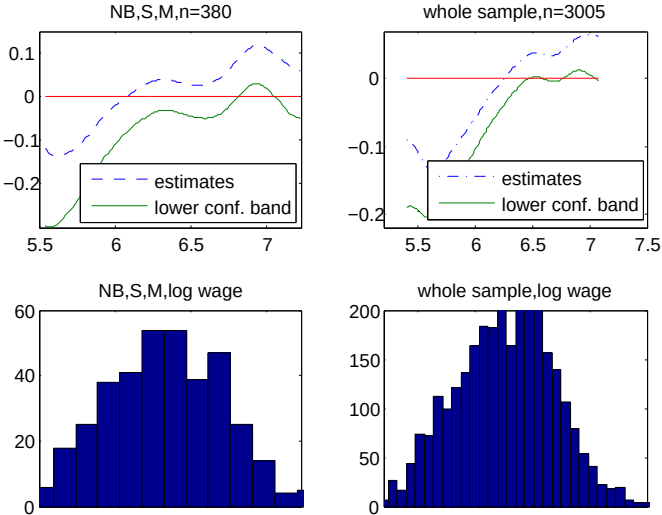
"College proximity"

Table: College proximity, local method (clrtest)

	NB,NS,NM	NB,NS,M	NB,S,NM	NB,S,M	B,NS,NM	B,NS,M	B,S,NM	B,S,M	All
Obs.	429	1191	307	380	5	138	314	246	3005
90%	NR	NR	NR	R	n.a.	NR	NR	NR	R
95%	NR	NR	NR	R	n.a.	NR	NR	NR	R
99%	NR	NR	NR	R	n.a.	NR	NR	NR	R

"R" stands for rejection and "NR" stands for no rejection.

“College proximity”



Parametric regression

$$\begin{cases} \theta(\cdot,1) \equiv \mathbb{E}[\underbrace{p_1 D(1-Z) - p_0 DZ}_{L^1} | Y, X1, X2, X3] \\ \theta(\cdot,0) \equiv \mathbb{E}[\underbrace{p_0 1(1-D)Z - p_1(1-D)(1-Z)}_{L^0} | Y, X1, X2, X3] \end{cases}$$

Table: Application results: Parametric (clrtest)

AE1998 same sex			Angrist1991 lottery			Card1993 proximity		
10%	5%	1%	10%	5%	1%	10%	5%	1%
NR	NR	NR	R	R	R	R	R	R

"R" stands for rejection and "NR" stands for no rejection.

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